

User Stories — Client Onboarding & KYC Workflow Optimization

US-01

As a **low-risk applicant**, I want my account activated immediately after document verification, so that I don't experience unnecessary delay for a profile that poses no meaningful risk.

Acceptance criteria: Given a computed risk_tier = "Low" and no PEP/sanctions match, when verification completes, then the system activates the account without routing to any Compliance review step, and logs the routing decision per FR-011.

US-02

As a **Compliance Lead**, I want every High-risk or PEP-flagged applicant automatically routed to Enhanced Due Diligence, so that the institution never relies on a human to remember to apply extra scrutiny.

Acceptance criteria: Given a PEP/sanctions match OR risk_tier = "High", when routing occurs, then the applicant is sent to the EDD path (FR-008) with two-reviewer sign-off required, regardless of any other factor.

US-03

As a **Medium-risk applicant**, I want a single standard review rather than the full enhanced process, so that my friction is proportional to my actual risk level.

Acceptance criteria: Given risk_tier = "Medium" and no override, when routing occurs, then exactly one standard Compliance review step is applied — not zero, not EDD.

US-04

As an **applicant with poor camera quality**, I want a clear prompt to retake my document photo rather than an opaque rejection, so that I understand what went wrong and can fix it myself.

Acceptance criteria: Given verification_match_rate < 40, when the system evaluates the upload, then the applicant receives a specific re-upload prompt (not a generic error), capped at 2 attempts per FR-009.

US-05

As an **applicant with an expired ID**, I want to be told immediately that my document is invalid, so that I don't waste time on a verification step that was doomed from the start.

Acceptance criteria: Given an ID with an expiry date in the past, when uploaded, then the system rejects it before running any match-rate calculation (FR-010) and prompts for a valid document.

US-06

As a **Product Manager**, I want to see Time-to-Onboard broken out by risk tier, so that I can demonstrate the fast-track is actually faster, not just theoretically faster.

Acceptance criteria: Given completed onboarding records across all tiers, when the TTO-by-tier query runs (Day 5), then Low-tier median TTO is measurably lower than Medium/High-tier median TTO, using real timestamps from the mock dataset.

US-07

As an Engineering Lead, I want an explicit precedence order when multiple rules apply to one applicant, so that I don't have to guess which rule wins during implementation.

Acceptance criteria: Given an applicant who triggers both FR-009 (failed verification) and FR-005 (PEP match), when the system processes them, then FR-009 resolves first per FR-014, and EDD routing is confirmed only after input quality is resolved.

Edge case (worked example, per FR-014): Given an applicant with a confirmed PEP match AND a `verification_match_rate` of 32 (below the FR-009 threshold of 40), when the system processes the application, then:

1. The system does not route to EDD immediately, despite the PEP match.
2. The system first prompts re-upload per FR-009, capped at 2 attempts.
3. If a passing match rate is obtained within 2 attempts, OR the attempt limit is exhausted and the applicant is escalated to manual review per FR-009's fallback, only then does the system apply FR-005.
4. The final destination is EDD in either case (since the PEP match alone guarantees it) — but the audit trail (FR-011) must show the re-upload step occurred *before* the EDD routing decision was logged, not after.

This is the FR-014 worked example treated as a concrete test case — it specifically guards against an implementation that short-circuits straight to EDD on seeing a PEP match and skips the input-quality gate.

US-08

As a **Compliance Lead**, I want a full audit trail for every applicant regardless of outcome, so that I can reconstruct and defend any routing decision to a regulator after the fact.

Acceptance criteria: Given any completed applicant record, when queried, then all FR-011 fields (inputs, score, tier, override flag, decision, timestamp) are present and non-null.

US-09

As an **applicant who matches a sanctions list**, I want my application routed to enhanced review rather than silently rejected, so that legitimate edge cases (e.g., name collisions) get human judgment rather than automated denial.

Acceptance criteria: Given a sanctions list match, when routing occurs, then the applicant is sent to EDD (not auto-rejected), per FR-005/FR-008.

US-10

As a **borderline-risk applicant** (score near the tier threshold), I want confidence that my tier assignment isn't arbitrary, so that I'm not penalized by a threshold that wasn't actually validated against real outcomes.

Acceptance criteria: Given the FR-004 threshold value, when finalized (Day 4), then it is backed by a documented precision/recall/F1 sweep across a wide threshold range (not just the original 70-vs-75 candidates), with the final choice (50) and the rejected F1-optimal alternative (40) both explained in terms of operational tradeoffs, not just statistical performance.

US-11

As an **Engineering Lead**, I want a defined fallback when the verification service itself fails (not the applicant's input), so that a system outage doesn't silently misclassify applicants as low-risk.

Acceptance criteria: Given a verification service failure, when the applicant's request is processed, then the applicant is held in pending_verification and never defaulted to risk_tier = "Low" by omission (FR-013).

US-12

As a **Product Manager**, I want the Verification Pass Rate reported by both risk tier and country, so that I can spot whether any country segment is producing unusually high verification failures (a possible UX or document-format issue, not necessarily a risk issue).

Acceptance criteria: Given the mock dataset, when the Day 5 SQL query runs, then pass rate is reported broken out by risk_tier × country, using real query output.

US-13

As a **Compliance Lead**, I want any future change to the risk threshold to require documented justification and my sign-off, so that the threshold can't drift without oversight after launch.

Acceptance criteria: Given a proposed threshold change, when submitted, then it must include a false-positive/false-negative impact analysis (FR-012) and is blocked until Compliance approves it.